

Week 6 Worksheet

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Exercise 1. The hamiltonian for the harmonic oscillator is

$$H = \frac{1}{2m}[p^2 + (m\omega x)^2].$$

Show that

$$H = \hbar\omega(a^\dagger a + 1/2),$$

where

$$\begin{aligned}a^\dagger &= \alpha x + i\beta p \\ a &= \alpha x - i\beta p.\end{aligned}$$

That is, determine α and β .

Exercise 2. A particle of mass m moves in the potential

$$V(x) = \begin{cases} +\infty, & x \leq 0 \\ \frac{1}{2}m\omega^2 x^2, & x > 0 \end{cases},$$

a harmonic well with a rigid wall slammed down at the origin. (This models, for example, a molecule vibrating against a hard surface, and the same equation governs the $\ell = 0$ radial motion of the three-dimensional oscillator.) Use the analytic solution of the full oscillator from lecture, whose normalized eigenstates are

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}, \quad \xi \equiv \sqrt{\frac{m\omega}{\hbar}} x,$$

with energies $E_n = (n + \frac{1}{2})\hbar\omega$ and Hermite polynomials are defined by: $H_0 = 1$ with a recurrence relation $H_{n+1}(x) = 2xH_n(x) - H'_n(x)$.

- a) Because $V = \infty$ for $x \leq 0$, the wave function must vanish there, in particular $\psi(0) = 0$. Argue that the allowed stationary states of the half-oscillator are precisely the eigenstates of the *full* oscillator that vanish at the origin, and identify which n these are. Hence write the allowed energies E_k ($k = 0, 1, 2, \dots$).

Hint: One approach is to use the result of Exercise 1 and the fact that $\psi_n(x) = (a^\dagger)^n \psi_0(x)$.

- b) Write the normalized ground-state wave function of the half-oscillator explicitly, in terms of ξ and H_1 . (Watch the normalization: the state now lives only on $x > 0$.)
- c) Find the most probable position of the particle in this ground state.
- d) The full oscillator has ground-state energy $\frac{1}{2}\hbar\omega$, but the half-oscillator's is $\frac{3}{2}\hbar\omega$. Explain physically why walling off $x < 0$ raises the zero-point energy.

Exercise 3. Bonus. Continue in the setup of Exercise 1.

- a) Show that $[a, a^\dagger] = aa^\dagger - a^\dagger a = 1$.
- b) Show that $[a^\dagger a, a] = -a$.
- c) If $N = a^\dagger a$ and $N\psi = \lambda\psi$ for some scalar λ and some normalizable ψ , show that $Na\psi = (\lambda - 1)a\psi$.
- d) Show that if ψ is normalized, then

$$\int |a\psi|^2 dx = \int \left(\psi^* (\alpha x)^2 \psi + \psi^* (\beta p)^2 \psi \right) dx - 1/2.$$

Hint: Use integration by parts, noting that $(|\psi|^2)' = (\psi^*)'\psi + \psi^*\psi'$.

- e) Show that there is a normalized solution ψ_0 to the TISE such that $a^\dagger a\psi_0 = 0$.
Hints: Use part (e) to show that $\int |a\psi|^2 dx = \int \psi^* N\psi dx$. Together with part (d), what does this tell you about repeated applications of a to ψ ?
- f) What is the energy of ψ_0 ?
- g) Show that $N = a^\dagger a$ is the **number operator**: If $\psi_n = (a^\dagger)^n \psi_0$ is the n^{th} stationary state, then

$$N\psi_n = n\psi_n.$$

- h) Show that if $N\psi_0 = 0$, then a complete set of normalized solutions to the equation $N\psi = \lambda\psi$ is given by $(a^\dagger)^n \psi_0$, so that $(a^\dagger)^n \psi_0$ are also a complete set of solutions to the TISE.